



A conceptual and statistical framework for delineating the timing of a stratigraphic transition: Holocene-Anthropocene boundary as a case study

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Abstract

The AWG (Anthropocene Working Group) proposed 1952 CE as the onset of the Anthropocene, marked by the detonation of the first hydrogen bomb, and designates Crawford Lake's sediments in Canada as the potential stratotype. While the proposal for the Anthropocene as an epoch has been rejected by the SQS (Subcommission on Quaternary Stratigraphy), ongoing discussions regarding this debate remain unresolved. Here, we explore the possible biases and pitfalls of selecting the onset date from a singular site as the GSSP (Global boundary Stratotype Section and Point) for a chronostratigraphic boundary, using the potential Holocene-Anthropocene boundary as an example. We address the challenges of identifying changes in anthropogenic markers, along with the inherent measurement and age uncertainties, by employing a multilevel Bayesian change point analysis integrated with an errors-in-variables approach applied to proxy data. Our analysis utilizes plutonium isotopes ($^{239+240}\text{Pu}$), SCPs (spheroidal carbonaceous fly-ash particles), and radiocarbon (F^{14}C) records from multiple locations established to evidence global anthropogenic imprints. Key findings include the identification of a population-level change point for $^{239+240}\text{Pu}$ at 1953 CE (1951.5–1954.5). While the standard practice for defining the onset date of a stratigraphic boundary typically involves selecting the date from the stratotype, our results suggest that 1953 CE should be the preferred date to define the onset of a potential Anthropocene Epoch. However, 1952 CE, which coincides with the first H-bomb test, cannot be statistically rejected as an alternative. Geographical clustering of SCPs data highlights regional variations in proxy signatures, while F^{14}C results emphasize a lag of 5.5–9 years corresponding to carbon residence time. Our research contributes to discussions surrounding the delineation of a potential

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boundary between the Holocene and Anthropocene Epochs. Additionally, it provides methodological insights into identifying change points in multiple datasets and temporal lags in the environmental impacts of human activities.

Keywords

Anthropocene, hierarchical Bayesian change point analysis, errors-in-variables, total evidence, international chronostratigraphic chart, radiocarbon atmospheric residence

Introduction

The proposal by the AWG to designate the onset of the Anthropocene as a new ongoing epoch for inclusion in the ICC (International Chronostratigraphic Chart) has recently been rejected by the SQS (IUGS, 2024). However, ongoing debates about the inclusion of the Anthropocene epoch in the ICC are far from settled (Adam, 2024; Rodriguez, 2024; Voosen, 2024; Witze, 2024a; Witze, 2024b). In this context, it is highly probable that a revised proposal may be re-submitted before long. Given this, we aim to contribute to these discussions by offering insights on the timing of a geological boundary for the Anthropocene, utilizing well-dated and measured data through its results and methodological approach.

The suggested onset of the Anthropocene Epoch in 1952 CE¹ by the AWG follows the paradigm described by established protocols (Head and Gibbard, 2015; Remane et al., 1996; Waters et al., 2023b). While different proposed events and various stratigraphic markers were considered for the onset of the Anthropocene (Waters et al., 2016, 2023a), the AWG decided to designate the mid-20th century as its start. The primary sedimentary marker chosen is plutonium fallout from thermonuclear bomb detonations², which can be measured worldwide (Waters et al., 2015, 2023a; Zalasiewicz et al., 2017a). A special issue dedicated to tracing human impacts through multiple proxy data from worldwide locations has been published (Waters et al., 2023b). Ultimately, a sediment core from Crawford Lake was selected as the GSSP candidate for the Anthropocene Epoch (Waters et al., 2023a; Witze, 2023). In this study, we challenge the start date of the Anthropocene using the data published in the special issue, along with additional plutonium data for Crawford

Lake provided by McCarthy et al. (2024). To achieve this, we employ multilevel (or hierarchical) Bayesian change point analyses that incorporate all available relevant data, while accounting for any reported uncertainties within the datasets.

From a historical perspective, many past scientific figures anticipated the idea of the Anthropocene, including Antoine Stoppani, Vladimir Vernadsky, Charles Darwin, and perhaps even Immanuel Kant (Guillaume, 2014; Kant, 1802/2012; Luciano and Zaroni, 2023; MacDuffie, 2022). However, their focus was primarily on human impact on the environment at smaller scales, rather than addressing changes to the earth system as a whole. Therefore, it is questionable whether these names truly pioneered the Anthropocene concept as we understand it today (Hamilton and Grinevald, 2015).

The current narrative of the Anthropocene originated with the seminal paper by Crutzen and Stoermer (2000). This paper proposed that the earth is in a new epoch, namely, the Anthropocene, that is, driven by extensive human-induced global earth system changes. Initially introduced as a modest yet cautionary proposal, their ideas sparked numerous debates over time, including the question of when the Anthropocene truly began. While the original proposal suggested the onset coincided with the Industrial Revolution (Crutzen, 2002; Crutzen and Stoermer, 2000), this idea then evolved (Crutzen and Steffen, 2003). Building on the views presented in Steffen et al. (2004), a more multilayered and data-based conception to draw the boundaries of a probable Anthropocene Epoch developed within a few years. According to this conception, the “Great Acceleration,” a period of dramatic human activity increase at the very beginning of the second half of the 20th century, is strongly linked to the onset of the

Anthropocene (Steffen et al., 2007). Since then, disagreements have arisen within the scientific community. Some argue that the Anthropocene began with the advent of large-scale agriculture (Fuller et al., 2011; Ruddiman, 2007), while others point to the extinction of numerous large mammals in the late Pleistocene (Doughty et al., 2010). Other candidates for the onset of the Anthropocene were the formation of anthropogenic soils (Certini and Scalenghe, 2011) and the *conquistadores* invasions of the Americas in the early 16th century (Lewis and Maslin, 2015).

In 2009, the ICS (International Commission on Stratigraphy) tasked a group of scientists known as the AWG with examining whether the Anthropocene merits formalization as a new epoch (Zalasiewicz et al., 2017a). According to the indicative voting in 2016 and the binding voting in 2019, the AWG announced that the boundary between the Holocene and the Anthropocene should be set at the mid-twentieth century (Waters et al., 2023a; Zalasiewicz et al., 2017a). This decision confirms the stratigraphic reality of the Anthropocene and recommends its formalization.

These decisions were made based on majority member votes but were met with criticism (Finney, 2014; Finney and Edwards, 2016). Taking into account the critiques, the AWG presented several arguments to justify their choice of 1950 CE as the onset of the Anthropocene (among others, Waters et al., 2016; Zalasiewicz et al., 2017b, 2024). However, their arguments did not completely settle the debate within the scientific community (Nielsen, 2021; Swindles et al., 2023). The AWG acknowledges the traceability of anthropogenic effects extending into the Holocene, as emphasized by their critics. However, none of these events matched the impact, synchronicity, and global scale of the “Great Acceleration,” claimed to have commenced around 1950 CE (Head, 2019). Consequently, they maintain that a mid-twentieth-century boundary level is stratigraphically optimal (Zalasiewicz et al., 2015). The call for the search for stratigraphic markers (Waters et al., 2018) resulted in a special issue (Waters et al., 2023b), culminating in the selection of a sediment core sample from Crawford Lake, Canada (McCarthy et al., 2023, 2024), as the most suited

GSSP candidate for the Holocene-Anthropocene transition (Waters et al., 2023a). While the AWG acknowledges that human-induced environmental changes began well before the mid-20th century, they argue that the stratigraphic signatures of the mid-20th century are either novel compared to those found in preexisting epochs or that the variability of signatures for many proxies already present in the Holocene has intensified, with this intensification continuing to accelerate (Waters et al., 2016). However, the rejection of the Anthropocene’s formalization by the SQS was primarily based on the diachronous nature of human impact (Gibbard et al., 2022; IUGS, 2024; Maslin et al., 2024; Walker et al., 2024; Witze, 2024a).

The AWG selected the plutonium isotopes, specifically ^{239}Pu and ^{240}Pu , as the primary stratigraphic marker (Waters et al., 2023a). The selection of $^{(239+240)}\text{Pu}$ is justified by its undeniable anthropogenic increase due to atomic bomb detonations, its immobility, long half-life, and its consistent spatial and temporal distribution across various global environments. In addition to $^{(239+240)}\text{Pu}$ isotopes, the special issue discussed many other proxies, highlighting SCPs, F^{14}C and nitrogen isotope ($\delta^{15}\text{N}$) as essential auxiliary markers (see Figure 2 in Waters et al., 2023b).

Historically, the selection of the onset date of a stratigraphic boundary depends on the dating of the GSSP. For example, the onset of the Holocene Epoch is defined through the age model of the NGRIP2 ice core (Walker et al., 2009), the Greenlandian Stage through NGRIP1, and the Meghalayan Stage by a speleothem sample from Mawmluh Cave (Walker et al., 2019). Similarly, the onset date of the Anthropocene is proposed to be defined through the CRA23-BC-1F-A sediment core from Crawford Lake, at the point where the initial major increase in $^{(239+240)}\text{Pu}$, described as the largest proportional increase between adjacent annual samples, is recorded (Waters et al., 2023a). This date also coincides with the first thermonuclear detonation on November 1, 1952 CE.

In the context of discussions surrounding the Anthropocene Epoch, we propose a reconsideration of selecting the onset date based on data from a single site, as this approach may have limitations similar to those of individual datasets from any single site. As highlighted by Ioannidis (2005), research findings

from a singular study are more likely to be false than those from a low-bias meta-analysis. Moreover, Ioannidis (2005: 700) aptly states, “[I]t is misleading to emphasize the statistically significant findings of any single team. What matters is the totality of the evidence.” We conduct a meta-analysis based on a multilevel Bayesian change point model. Multilevel Bayesian models applied in meta-analyses are intriguing due to their ability to express various uncertainties in the model, including measured proxy and age uncertainties, and to share information between each parameter within the model (Kruschke and Liddell, 2018a).

Bayesian change point analysis (Gallagher et al., 2011) has been applied in previous studies within the earth sciences domain. For instance, it has been utilized to identify change points in global temperatures during the 20th century (Cahill et al., 2015b) and to assess the credibility of events such as the 4.2 ka BP event (Ön, 2023a). Additionally, different Bayesian approaches for a set of measured quantities from the same locality have been used to detect change points in some papers from the special issue, specifically Kaiser et al. (2023) and Thomas et al. (2023). Furthermore, Bayesian models incorporating an errors-in-variables approach, integrated with Gaussian processes, have been employed to model sea level changes (Cahill et al., 2015a). Multilevel Bayesian approaches have also been instrumental in temperature reconstruction studies (e.g., Tingley and Huybers, 2010; Ön et al., 2023).

In our study, we employ an integrated change point approach that is both multilevel and incorporates errors-in-variables, akin to the methodology proposed by Ön (2023b). However, our multilevel approach distinguishes itself by utilizing precisely dated and measured datasets under entirely different model assumptions, while also accounting for measurement uncertainties in both proxy and dating.

Our results, derived from multiple $^{(239+240)}\text{Pu}$ datasets, do not rule out 1952 CE as a potential start date for the Anthropocene Epoch when $^{(239+240)}\text{Pu}$ is considered the primary marker. However, we propose that 1953 CE offers a more accurate estimate at present, given the residence time of Pu in various reservoirs. This estimate may be revised as new data emerges and is assimilated into the model. Additionally, the results on F^{14}C reflect the lag for carbon

residence time in the reservoirs. The variable results of SCPs data for each dataset mirror the geographical emission patterns.

Data

In this study, we reviewed 12 different GSSP candidate sites featured in the dedicated special issue (Waters et al., 2023b). However, not all the data presented in this special issue proved to be suitable candidates (Borsato et al., 2023; Himson et al., 2023; Wagreich et al., 2023). Therefore, we utilized nine sites with datasets that meet the criteria to serve as a stratotype (Figure 1).

In our statistical approach, specifically multilevel Bayesian change point analysis, we assume that all datasets share the same structural pattern, meaning the response of the selected proxy should be globally similar. While many measurements are presented in the special issue (Table 3 in Waters et al., 2023b), the majority of these proxy data exhibit regional variations. However, among the measured proxies, the $^{(239+240)}\text{Pu}$ and F^{14}C datasets clearly demonstrate the effects of nuclear tests and display a similar pattern: an almost constant level followed by an abrupt increase (Figure 2(a) and (c)). Additionally, the increase in electricity demand and the introduction of fuel-oil combustion, alongside coal, yield a comparable pattern in the SCPs data across all measured localities (Figure 2(b)).

However, not all available data were incorporated into our primary analyses, as outlined in Table 1. Specifically, the $^{(239+240)}\text{Pu}$ datasets from North Flinders Reef (Zinke et al., 2023) and the Palmer ice core (Thomas et al., 2023) were excluded, along with the SCPs datasets from the Palmer ice core and Searsville Lake (Stegner et al., 2023), as well as the F^{14}C dataset from Sihailongwan Lake. The exclusion of the mentioned $^{(239+240)}\text{Pu}$ datasets is based on the following reasons: (i) Insufficient data available before 1952, with only one measurement for North Flinders Reef and three measurements for the Palmer ice core. (ii) Relatively low temporal resolutions of the datasets, with measurements existing for every 3.4 years for North Flinders Reef and 2.3 years for the Palmer ice core. (iii) Post-1952 data exhibit high variability (see Figure 2(a)), where a multilevel analysis yields a change point value exceeding what is deemed reasonable.

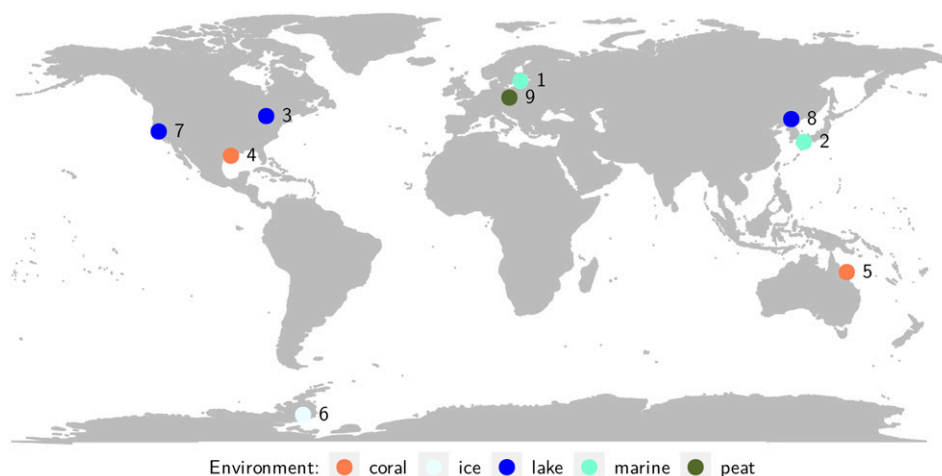


Figure 1. The geographical locations of the datasets used in the present study. Refer to Table 1 for the corresponding references of each dataset. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

The SCPs dataset from Searsville Lake exhibits considerable variability after the initial rise (Figure 2(b)). This pronounced variability, along with the absence of a clear persistent rise observed in other SCPs datasets, is attributed to elevated sedimentation rates (Stegner et al., 2023). The presence of non-zero data points between densely populated zeroes in Palmer ice core SCPs dataset poses a significant challenge for unbiased transition point identification in change point analysis (Figure 2(b)). Consequently, these datasets were not further considered for the purpose of the present study.

We also excluded the reported $F^{14}C$ data derived from sediment soot in Sihailongwan Lake, as it was identified primarily as a proxy for fossil fuel emissions, with some influence from global nuclear weapon tests, and is not comparable to the $F^{14}C$ datasets from the other studies (Han et al., 2022, 2023).

Moreover, some studies have reported measurements from different cores, such as $F^{14}C$ data from Crawford Lake and Beppu Bay. Using measurements from multiple cores would introduce additional uncertainty into the analysis; thus, we selected the core with the most complete dataset (Table 1).

However, the published $(^{239+240})Pu$ data from Crawford Lake in the special issue was derived from core CRA19-2FT-D1 (McCarthy et al., 2023), while

the proposed GSSP core was identified as CRA23-BC-1F-B (McCarthy et al., 2024; Waters et al., 2023a). The primary issue with the $(^{239+240})Pu$ measurements from CRA23-BC-1F-B is that they begin in 1949 CE, resulting in insufficient data coverage before 1952. Therefore, for the Crawford Lake $(^{239+240})Pu$ dataset, we decided to incorporate all available measurements from CRA19-2FT-D1, CRA22-1FT-3, and CRA23-BC-1F-B, applying the latest age models presented by McCarthy et al. (2024). We believe that the uncertainty introduced by utilizing multiple cores will be minimal, as the age models for all these cores are based on varve counting, which is corroborated by chronostratigraphic markers (Lafond et al., 2023; McCarthy et al., 2023, 2024).

Methods

Data preprocessing

The datasets were confined to the interval between 1900 and 2000 CE. While some proxy and age errors were provided by the original authors as one standard deviation, others were specified as upper and lower limits around a mean value. In cases where limits were provided, we calculated the average distance from the limits to the mean value to determine the error value.

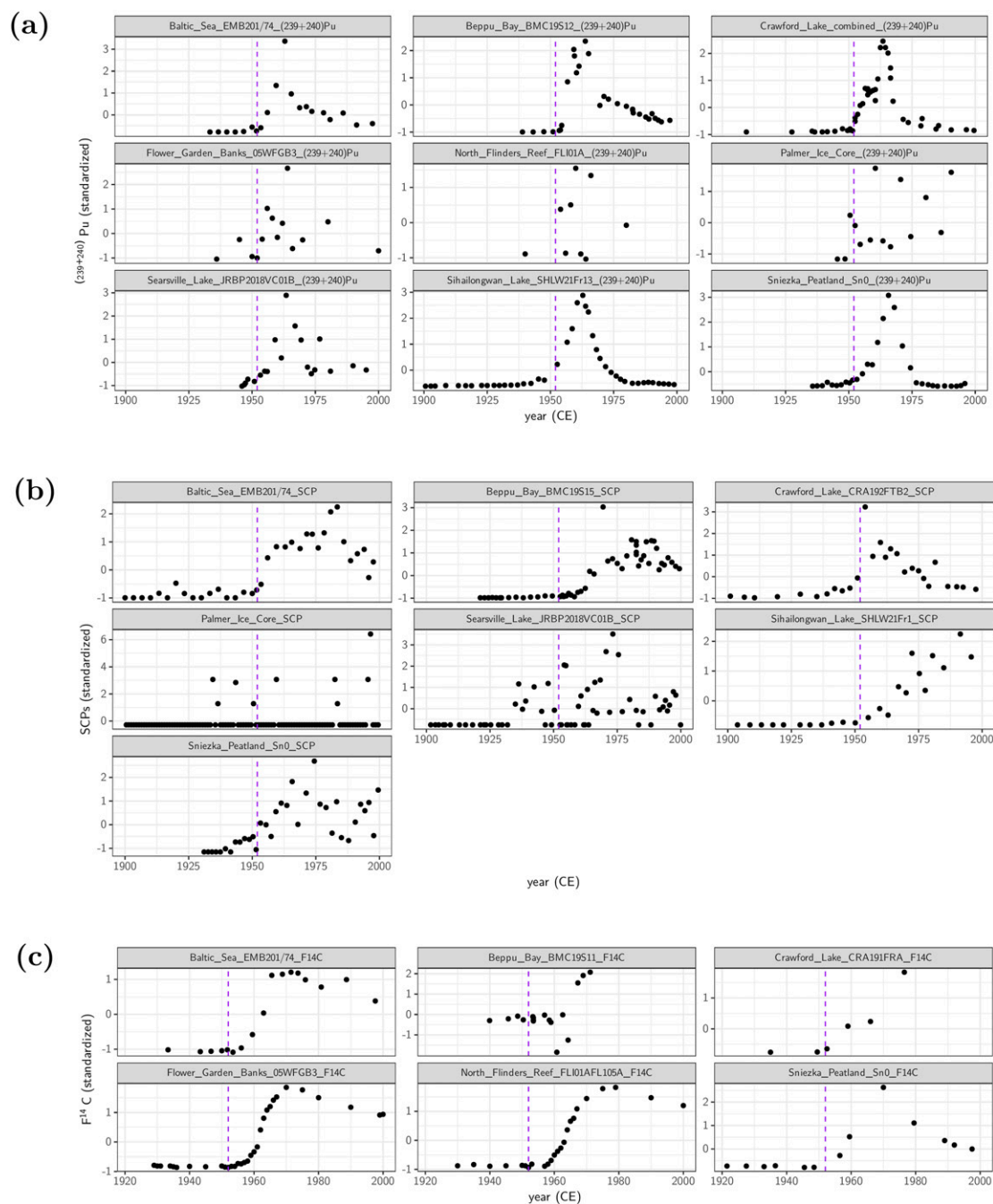


Figure 2. (a) $(^{239}+^{240})\text{Pu}$ (b) SCPs (c) F^{14}C datasets analyzed here for the 20th century interval. The purple dashed lines indicate the year 1952 CE. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

Table 1. List of datasets presented in this study. If a dataset is used, the name of the sediment core is indicated; otherwise, “—” denotes that the dataset is not available, and “x” denotes that the dataset is available but not used.

Location	$(^{239+240}\text{Pu})$	SCPs	F^{14}C	Reference
1. Baltic Sea	EMB201/7-4	EMB201/7-4	EMB201/7-4	Kaiser et al. (2023)
2. Beppu Bay	BMC19 SI-2	BMC19 SI-5	BGI19 SI-1	Kuwaie et al. (2023)
3. Crawford Lake	CRA19-2FT-D1	CRA19-2FT-B2	CRA19-2FT-IFR-A	McCarthy et al. (2023, 2024)
4. Flower Garden Banks	05WVFGB3	—	05WVFGB3	DeLong et al. (2023)
5. North Flinders Reef	x	—	FLI01 A	Zinke et al. (2023)
6. Palmer ice core	x	x	—	Thomas et al. (2023)
7. Searsville Lake	JRBP2018-VC01 A	x	—	Stegner et al. (2023)
8. Sihailongwan Lake	SHLW21-Fr-13	SHLW21-Fr-1	x	Han et al. (2023)
9. Śnieżka Peatland	Sn0	Sn0	Sn0	Flakiewicz-Kozieł et al. (2023)

Additionally, SCPs data from Beppu Bay and Sihailongwan Lake were reported without specific measurement errors. For these datasets, we adopted a 10% uncertainty level of the measurement data, as approximately reported in Rose (2008), to represent the proxy error in the model.

One data point from Searsville Lake SCPs measurements, along with six data points from Flower Garden Banks $(^{239+240}\text{Pu})$ datasets, were reported without any proxy measurement uncertainty. To address this, we imputed these unavailable values randomly using a half-normal distribution, where the distribution has a mean of zero and a standard deviation equal to 0.05 times the proxy measurement.

Following the imputation process, each dataset was standardized by subtracting the mean of the dataset and dividing by the standard deviation of the dataset. Measurement errors were standardized by dividing them by the standard deviation of the measurement dataset.

As our primary interest was to detect the start of the permanent increase in the datasets as the change point, we implemented distinct change point scenarios for each dataset. We incorporated slight variations in the regression models to emphasize the identification of the second change point in each data series using the `mcp` package (Lindeløv, 2020) in R (R Core Team, 2021). The best-fitting model among these various change points was determined through a combination of WAIC (widely applicable information criteria) (Watanabe, 2010) and leave-one-out cross-validation techniques, implemented using the `loo` package (Vehtari et al., 2017) in R. These singular analyses were conducted on datasets that show a second change in the series, including the Baltic Sea, Beppu Bay, Crawford Lake, Searsville Lake, Sihailongwan Lake, and Śnieżka Peatland $(^{239+240}\text{Pu})$ datasets, as well as the Baltic Sea and Beppu Bay SCPs datasets, and the Baltic Sea, Flower Garden Banks, North Flinders Reef, and Śnieżka Peatland F^{14}C datasets.

Subsequently, values younger than the mean of the posterior distributions of the corresponding second change points were excluded. For the remaining data, the mean of the means of the posteriors for each proxy set was calculated, and data points younger than this value were removed from each dataset.

Lastly, as our model is an errors-in-variables model, measurement and age errors are incorporated as

precision, that is, the reciprocal of the variance. To ensure numerical stability, a minimal value of 10^{-10} is assigned to the error terms that are equal to zero.

Change point analysis

We employed a continuous piecewise linear regression model with a single change point, denoted by c , indicating the break point in the line. The assumption is a constant level followed by a ramp. This assumption is reflected in the following set of expressions, representing the model's likelihood:

$$y_i \sim N(\mu_i, \sigma_i^2), \quad (1)$$

where

$$\mu_i = I_{x_i \geq \min x}^* \alpha + I_{x_i \geq (c+c_k)^*}^* \beta X_i, \quad (2)$$

$$\sigma_i^2 = (\sigma_i^{\text{proxy}})^2 + \tau^2. \quad (3)$$

In eq. (2), I is the indicator function, and $\min x$ is the minimum date value in the dataset. The change point at the population level, c , follows a truncated t -distribution:

$$c \sim t_{v_{cp}}(\mu_{cp}, \sigma_{cp}^2) \text{ T}(\min x, \max x), \quad (4)$$

where $\max x$ is the maximum date value in the dataset. This truncation ensures that the change point falls within the specified interval. We employ the centered change points for each dataset, c_k^* , as:

$$c_k^* = c_k - \text{mean}\{c_k\}_{\forall k}, \quad (5)$$

$$c_k \sim t_5(0, \sigma_{cp_k}^2) \text{ T}(\min x - c, \max x - c). \quad (6)$$

By subtracting the average of the change points, the multilevel model centers the change points of each dataset around the population-level change point.

Following Gelman (2006), we define the scale parameters of the t -distributions using half-Cauchy distributions that are weakly informative:

$$\sigma_{cp} \sim t_1^+(0, 50^2), \quad (7)$$

$$\sigma_{cp_k} \sim t_1^+(0, 50^2). \quad (8)$$

We sample the location parameter for the change point distribution, μ_{cp} , from a uniform distribution:

$$\mu_{cp} \sim U(\min x, \max x). \quad (9)$$

As suggested by Kruschke (2015), the normality parameter (degrees of freedom), v_{cp} , is drawn from an exponential distribution:

$$v_{cp} \sim \text{Exp}(1/30). \quad (10)$$

We specify the parameters of the regression model using t -distributions that are non-informative:

$$\alpha \sim t_3(0, 5^2), \quad (11)$$

$$\beta \sim t_3(0, 100^2). \quad (12)$$

In eq. (2), X represents the local origin for the second segment and is defined as:

$$X_i = x_i^* - (c + c_k), \quad (13)$$

where c_k corresponds to the centered change point corresponding to x_i^* . We model the uncertainty in age measurements using normal distribution:

$$x_i \sim N(x_i^*, \sigma_{age}^2), \quad (14)$$

and assume that the unobserved x_i^* follows a uniform distribution that is non-informative:

$$x_i^* \sim U(\min x, \max x). \quad (15)$$

In eq. (3), σ_i^{proxy} denotes the reported proxy measurement errors, and τ accounts for micro-scale variation, which is sampled from a half-Cauchy distribution that is weakly informative:

$$\tau \sim t_1^+(0, 5^2). \quad (16)$$

The program JAGS (Plummer, 2003) was employed for Gibbs sampling through the `runjags` interface (Denwood, 2016) in R (R Core Team, 2021) to derive posterior distributions for the model. The initial 20 000 iterations were discarded as adaptation steps, followed by another 20 000 iterations as burn-in steps. For each dataset and change point

assumption, three separate MCMC (Markov chain Monte Carlo) chains were executed, each comprising 20 000 iterations.

Convergence of the Bayesian model was confirmed through the evaluation of convergence diagnostics for each model and dataset. This assessment entailed a thorough examination of trace plots, autocorrelation plots, Gelman-Rubin statistics, and density plots, across all three runs for all parameters. For detailed convergence diagnostic plots, please refer to the supplementary information (Figure S4, S5 and S6).

Simulation

In order to test the Bayesian change point model effectiveness, we simulated a dataset similar to the $^{(239+240)}$ Pu datasets consisting of randomly selected 20 data points using JAGS (Figure 3). The dates for these points are drawn from a uniform distribution spanning the years 1930 to 1970, and the date uncertainties are derived from a standard half-normal distribution. The y -values are sampled from a normal distribution with a standard deviation of 0.25. The means are determined by a piecewise linear function that mimics the $^{(239+240)}$ Pu datasets, introducing a change point at the year 1952:

$$\hat{\mu}_y = \begin{cases} 0 & \text{if } x \leq 1952 \\ 0.25(x - 1952) & \text{if } x > 1952 \end{cases} \quad (17)$$

The errors in the y -values are drawn from a half-normal distribution with a mean of zero and a standard deviation of 0.1.

We conducted change point analysis on the simulated data following the model structures outlined in subsection on change point analysis. However, it's important to note that the approach employed here is not multilevel; instead, it is tailored for individual datasets for demonstration purposes. Despite not being nested in a multilevel model, the analysis of the simulated data demonstrates the effectiveness of the model (Figure 3).

Additionally, we conducted simulations for multiple datasets and their corresponding multilevel model results to validate the efficacy of the model. Further details can be found in the supplementary information and through Figures S1, S2, and S3.

Results

Three distinct analyses have been conducted, and the results are presented in three sections. In the

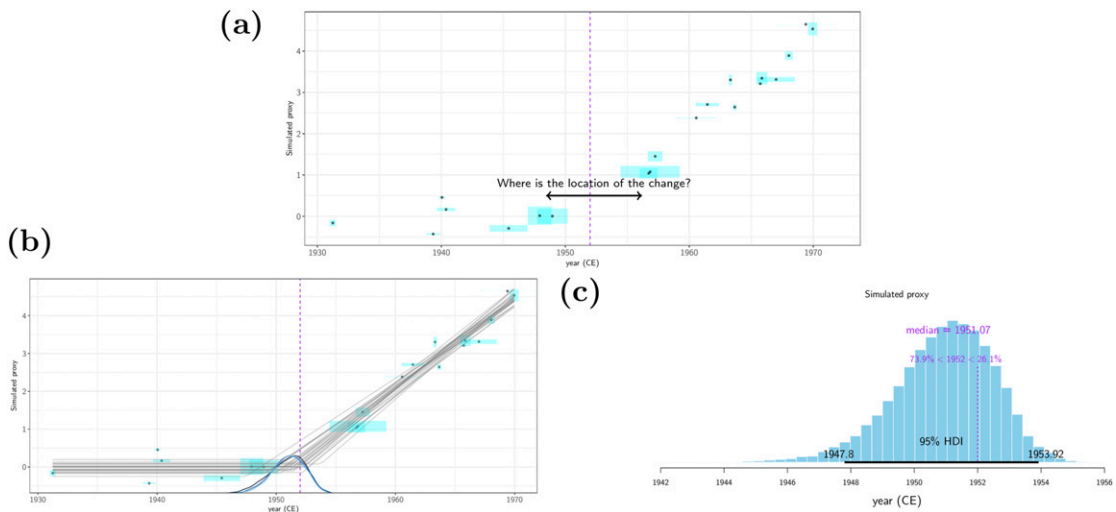


Figure 3. (a) Simulated data with cyan boxes representing one sigma uncertainty. (b) Analysis result on the simulated data. (c) Posterior distribution of the change point. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

following, the 95% HDI (highest density interval) is provided in parentheses after the median value.

The outcome of the multilevel change point analysis for the $(^{239+240}\text{Pu})$ datasets indicates a median date of 1953 CE (1951.5–1954.5) as the population-level change point (Figure 4). The results from each dataset are scattered around this median value (Table 2 and Figure 4). Specifically, the median date for the change point result of Sihailongwan Lake dataset reveals minimum values of 1948.5 CE (1947–1950). In contrast, the median date for Searsville Lake dataset shows the opposite extreme with a date of 1955.5 CE (1953–1958).

The results from the change point analysis of the SCPs datasets are relatively more scattered, but show regional clusters (Table 2 and Figure 5). The change point analysis for the Canadian Crawford Lake SCPs dataset yields a result of 1939 CE (1935.5–1943). In Europe, both the Baltic Sea and Śnieżka Peatland datasets exhibit a cluster of results with 1947 CE (1942.5–1951.5) and 1947.5 CE (1943–1952), respectively. Similarly, in Asia, Beppu Bay and Sihailongwan Lake datasets present a cluster of results with 1955 CE (1952.5–1957.5) and 1953 CE (1949–1957), respectively. The population-level change point corresponds to the date 1948.5 CE (1945.5–1951).

The results from the change point analysis of the F^{14}C datasets reveal a much more compact result compared to the $(^{239+240}\text{Pu})$ and SCPs results (Table 2 and Figure 6). The population-level change point is determined to be 1959.5 CE (1958–1961.5), and the results are clustered around this population-level change point. The only exception is the Śnieżka Peatland dataset with a change point at 1956.50 CE (1954–1959.5).

Discussion

In a change point analysis, the primary goal is to pinpoint the exact location of a change point. However, measurement data inherently exhibits mathematical discontinuities (Figure 2). When dealing with discontinuous data, there is no assurance that an apparent increase corresponds to the first actual increase. Additionally, the data might have undergone an increase (or decrease) in an unobserved

interval, leading to a potential misinterpretation when only considering the measured points.

Rather than pinpointing a specific data point as indicating a change, it is crucial to present a statistical uncertainty interval that encompasses a probable change point (Figure 3). A key strength of the data utilized in this study is the availability of multiple datasets that are precisely dated and measured, with explicitly reported measurement uncertainties. This abundance and precision in measurements, along with their associated uncertainties, enable a meta-analysis that integrates all relevant evidence. Such an approach can mitigate the potential flaws or biases that might arise from relying on a single study, thereby enhancing the reliability of the inference (Carnap, 1947, 1962). A Bayesian approach is particularly well-suited for this task, as it quantifies uncertainty by incorporating assumptions and prior information into probability distributions. This method provides a comprehensive framework for the nuanced quantification of uncertainty, especially in change point analysis, and more broadly, in any statistical analysis.

In light of this, the multilevel statistical meta-analysis approach applied in this paper is particularly useful when assuming the presence of a consistent global structural temporal pattern across the data, as is clearly observed in the Anthropocene markers. Within the framework of meta-analysis thinking, each finite dataset is considered a random sample contributing to the overall result rather than determining it (Cumming, 2014; Kruschke and Liddell, 2018a). This idea naturally leads to using a multilevel Bayesian approach, where different levels of data, represented as probability distributions, can inform each other. Such a configuration promotes the convergence of extreme low-level cases—results from individual localities—toward the statistical modes of the group (Kruschke and Liddell, 2018a, 2018b).

The primary focus of inquiry in this study is to determine the location of the change point. Our model proposes a line parallel to the time axis, resembling a plateau, as the initial segment of the dataset. This plateau is consistently followed by a segment with the same slope after the change point across all datasets. This observed pattern is evident in nearly all datasets (see Figures 4(a), 5(a), and 6(a)).

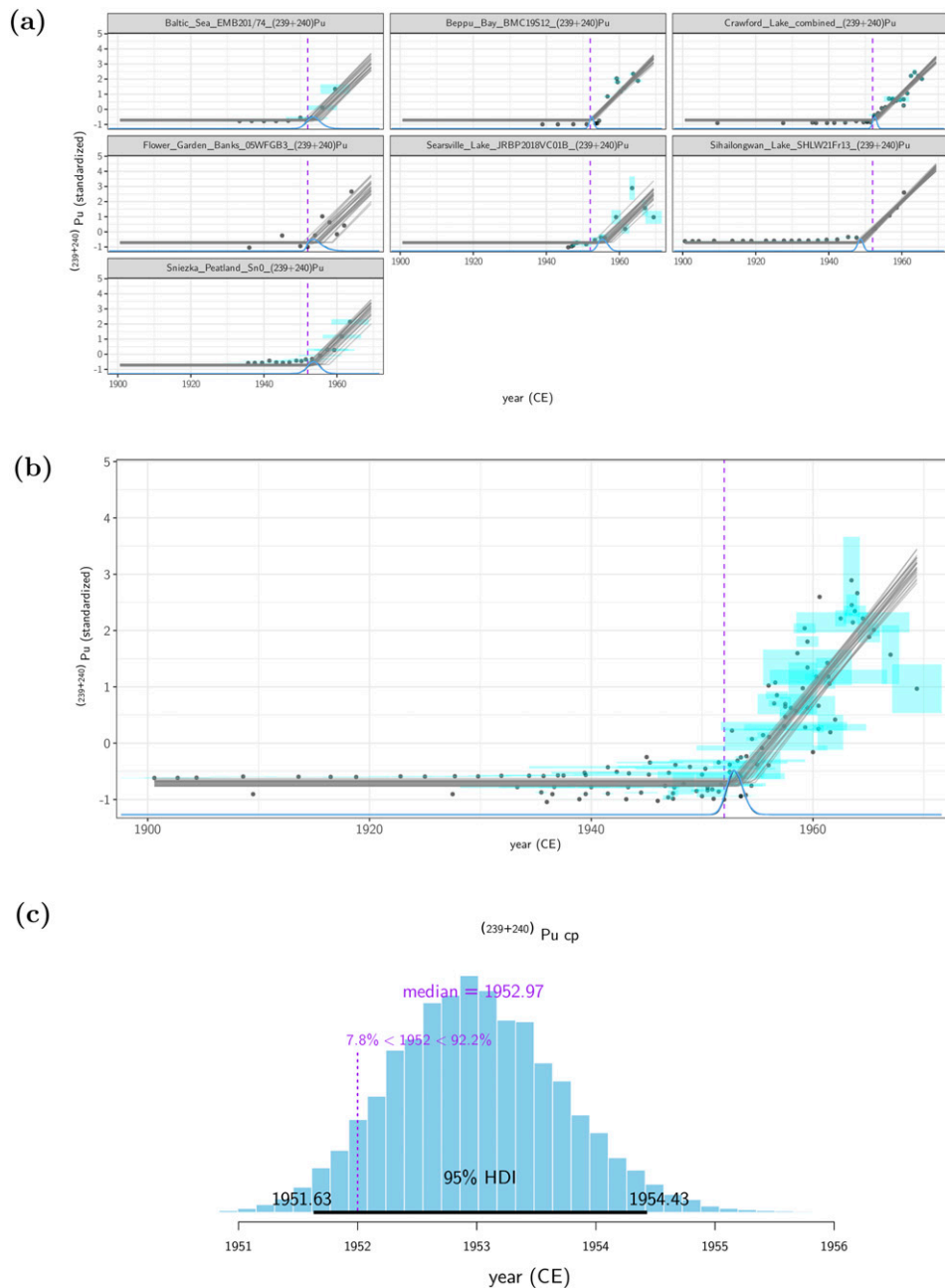


Figure 4. Multilevel change point results of $(^{239+240}\text{Pu})$ for (a) each dataset and (b) the population dataset (overall change point). The posterior distribution for the population dataset is shown (c). Cyan boxes in (a) and (b) represent one sigma uncertainty intervals associated with dates and measurements. Gray lines depict 25 random solutions generated from posterior draws. The purple dashed lines indicate 1952 CE, and the black bar below the posterior distribution in (c) indicates the 95% HDI. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

Table 2. Results of the change point analyses. The median and 95% HDI values (in brackets) of the posteriors of the change point for each proxy are listed.

	$(^{239+240})\text{Pu}$	SCPs	F^{14}C
1. Baltic Sea	1953.5 (1950–1957.5)	1947 (1942.5–1951.5)	1961 (1956.5–1966.5)
2. Beppu Bay	1952.5 (1951.5–1953.5)	1955 (1952.5–1957.5)	1961 (1959.5–1962.5)
3. Crawford Lake	1952.5 (1951.5–1953.5)	1939 (1935.5–1943)	1961.5 (1957.5–1964.5)
4. Flower Garden Banks	1954 (1951–1959)	—	1958 (1957–1959.5)
5. North Flinders Reef	—	—	1960.5 (1959–1962)
7. Searsville Lake	1955.5 (1953–1958)	—	—
8. Sihailongwan Lake	1948.5 (1947–1950)	1953 (1949–1957)	—
9. Śnieżka Peatland	1953.5 (1950.5–1957)	1947.5 (1943–1952)	1956.5 (1954–1959.5)
Population-level change points	1953 (1951.5–1954.5)	1948.5 (1945.5–1951)	1959.5 (1958–1961)

Given the discontinuity of the datasets at various points and the measurement uncertainties, particularly around suspected change points, articulating the change location necessitates a probability distribution to encapsulate the associated uncertainties of the change points (see [Figure 3](#)).

The AWG designated the year 1952 CE as the boundary for the Holocene-Anthropocene transition. This choice is linked to the first H-bomb test, namely, Ivy Mike, conducted on 1 November, 1952. While there were prior nuclear explosions between 1945 and 1952 CE, their effects appear to be only discernible in regional geological archives ([Waters et al., 2018](#)). Notably, atmospheric Pu has a relatively short atmospheric residence time, ranging from 70 days to a couple of years ([Waters et al., 2015](#)). According to [Hirose and Povinec \(2015\)](#), the average stratospheric mixing time is 1.5 ± 0.5 years and decreases exponentially. As indicated by [Stegner et al. \(2023\)](#), the lag can be approximated to be around 1 year.

Our analysis of the $(^{239+240})\text{Pu}$ datasets, excluding those discussed in section on data, reveals a population-level change point in 1953, with a 95% uncertainty interval of 1951.5–1954.5. This median date likely corresponds to the timing of the first H-bomb detonation and accounts for a lag of a couple of months for the global fallout of $(^{239+240})\text{Pu}$, which occurred on November 1, 1952.

The standard practice for selecting a chronostratigraphic boundary is to identify the first change in the physical properties or fossil content of

the strata at the stratotype ([ICS, 2024](#); [Remane et al., 1996](#)). While the peak of the $(^{239+240})\text{Pu}$ data occur later in the time series—likely aligning with the Partial Nuclear Test Ban Treaty of 1963—based on our assumptions and approach, the year 1953 CE, which marks the average start of the increase in multiple well-measured and well-dated datasets, is the most appropriate and statistically relevant stratigraphic horizon for defining the onset of the Anthropocene Epoch.

In a multilevel Bayesian approach, even when individual datasets do not converge, extreme cases are often pulled toward the modes of the group data. However, in a setting with only nine datasets, where two have low resolution or do not conform to the structural pattern of the model, there is a potential to skew the overall results. We conducted the same analysis using the complete $(^{239+240})\text{Pu}$ datasets, including data from the North Flinders Reef and the Palmer ice core, as shown in [Figure S7](#) and [Table S1](#). The results from these two sites are not consistent with the rest of the datasets, significantly influencing the overall outcome. While incorporating all relevant data is a central argument of this study, we chose to exclude the North Flinders Reef and Palmer ice core $(^{239+240})\text{Pu}$ datasets due to the reasons detailed in section on data and the findings presented in the [supplemental material](#). These datasets yield irrelevant median dates with large uncertainty intervals, which in turn affect the overall results to some extent.

A recent study aiming to confirm the distinct character of the mid-20th century increase in

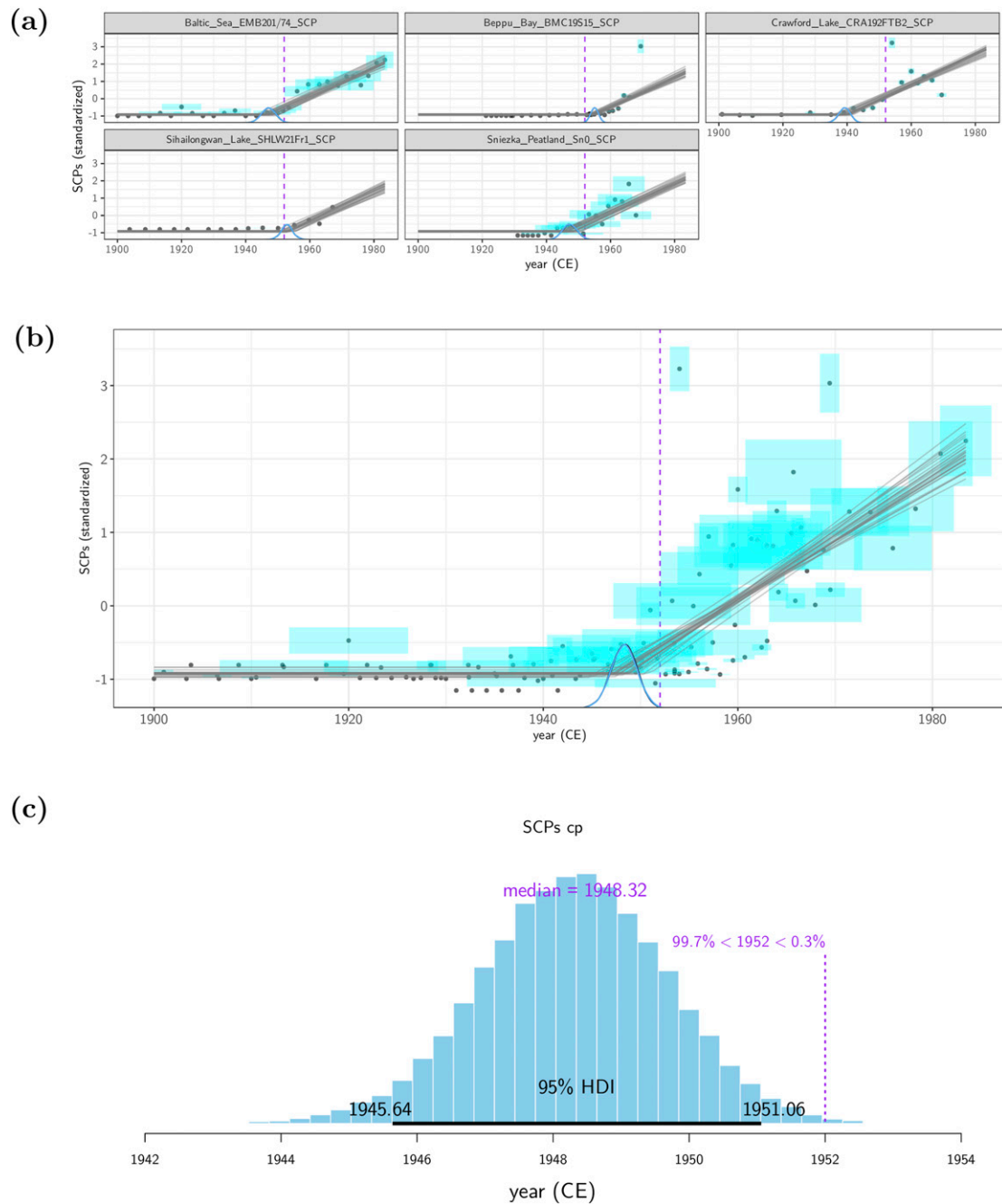


Figure 5. Same as Figure 4, but for the SCPs datasets. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

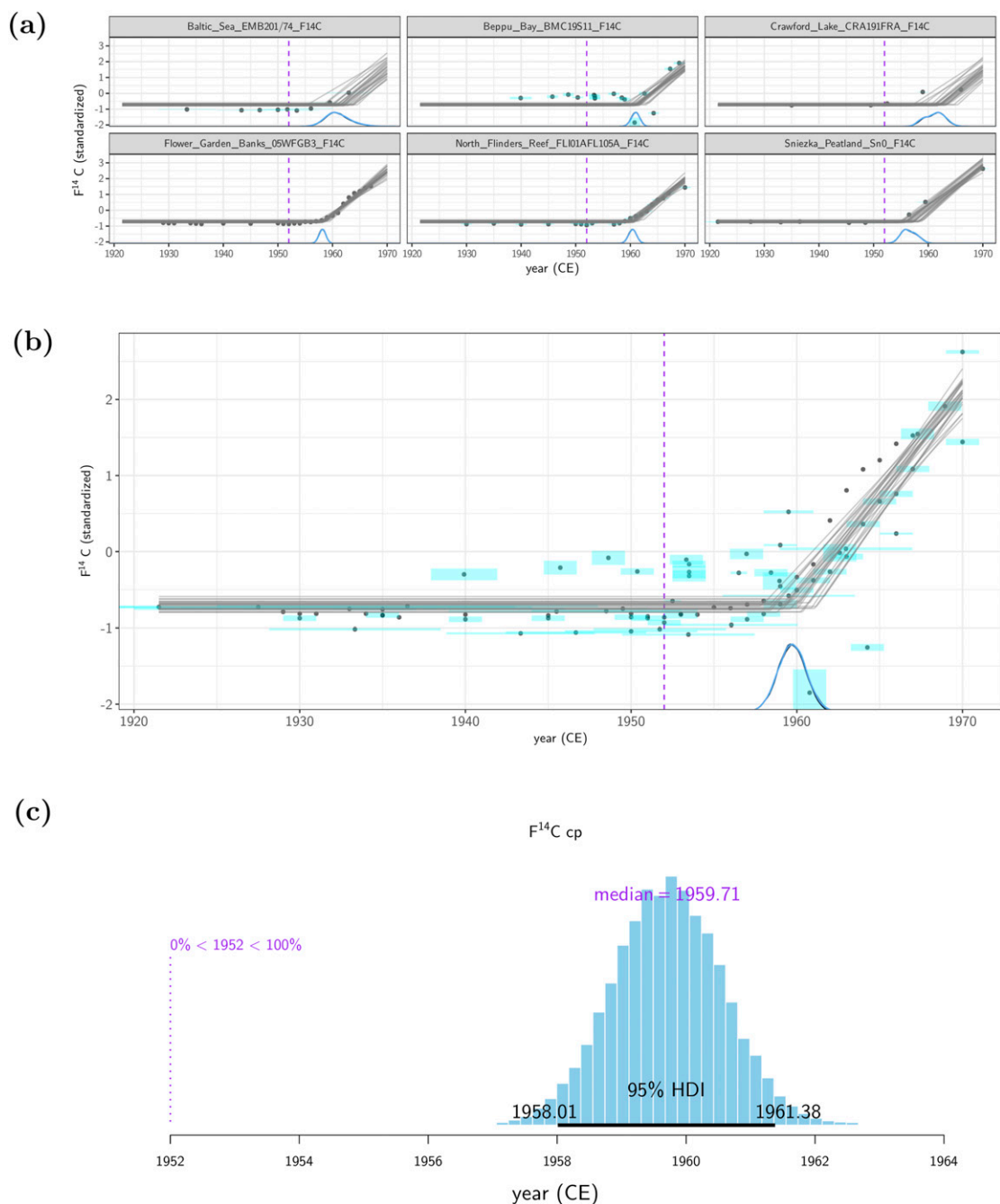


Figure 6. Same as Figure 4, but for the $F^{14}C$ datasets. For interpretation of the references to colours in this figure legend, refer to the online version of this article.

anthropogenic signatures analyzes a wide range of proxy types through a two-step non-Bayesian approach (Kuwaie et al., 2024). While their estimated onset points vary depending on specific settings, they ultimately suggest 1952 ± 3 as the onset of the “Great Acceleration,” a finding that supports the mid-20th century hypothesis and, to some extent, aligns with our results.

The SCPs results reveal a distinct geographical pattern. The change points of the datasets from Europe exhibit a clustering around the year 1947 CE (Fialkiewicz-Kozieł et al., 2023; Kaiser et al., 2023), while they cluster around 1954 CE in Asia (Han et al., 2023; Kuwaie et al., 2023). The change point of the dataset from North America (McCarthy et al., 2023), yields a median date of 1939 CE. Although the Searsville Lake SCPs dataset from North America was not analyzed here, an early increase in the mid-1930s is apparent (Stegner et al., 2023). Consequently, consistent with previously observed emission patterns (Rose, 2015; Ruppel et al., 2013), the variations in SCPs change points are geographically associated: an early increase in fossil fuel combustion in North America, followed by a rise in Europe due to high electricity demand and the introduction of fuel-oil combustion after World War II, and a later increase in Asia attributable to its industrial development.

The $F^{14}C$ datasets yield a more concise perspective compared to the $(^{239+240})Pu$ and SCPs datasets, with reduced standard deviation for the median dates of the posterior distributions. The median dates predominantly cluster around 1960 CE, except for the Śnieżka Peatland dataset. Previous studies have suggested that the introduction of atmospheric ^{14}C into surface waters and the atmosphere typically entails an approximate lag of 10 years for ^{14}C data to become detectable in shallow sediments, living shallow corals, or animals (Key, 2001; Waters et al., 2019). Assuming the initiation of the atmospheric ^{14}C increase corresponds to the first H-bomb test at the end of 1952 CE, the lag ranges from approximately 5 to 8.5 years for the marine and limnic datasets, considering the 95% uncertainty interval of 1958–1961.5. However, for the Śnieżka Peatland dataset, this lag appears to be only 4 years. Radiocarbon

measurements for Śnieżka Peatland were conducted on *Polytrichum strictum* stems with leaves, *Sphagnum* stems, and *Sphagnum russowii* stems (Fialkiewicz-Kozieł et al., 2023). The smaller lag time is likely attributed to the carbon capture of the peat mosses, primarily from the atmosphere (Blaauw et al., 2004). Indeed, artificial ^{14}C in the atmosphere and tree rings was first detected in 1955–1956 CE on a global scale (Hua et al., 2013). However, it is also possible that since the age model of the Sn0 core, whose data is used in this study, is based on ^{210}Pb dating and a comparison of measured radioisotopes to global fallout dates, the results of $(^{239+240})Pu$ and SCPs provide strong support for the robustness of the Sn0 age model.

The availability of precise measurements of the effect of the first H-bomb detonation and its reflection in multiple sediment datasets enables us to approximate the residence time of carbon in the atmosphere. Thus, to estimate the average reservoir effect of atmospheric carbon, we conducted the same procedure, excluding the ^{14}C data of the Śnieżka Peatland dataset. The median of the posterior distribution is 1960.5 CE, with a 95% uncertainty interval of 1958.5–1962 (Figure S7). This provides an estimate that atmospheric carbon takes 5.5 to 9 years to be incorporated into oceanic and limnic archives. Previous studies, such as the 5–10 year estimate by Nydal and Lövseth (1970), the 4-year estimate by Watson et al. (1990), and the approximately 5-year estimate by Levin and Hesshaimer (2000), align with our findings. The presence of a lag was noted by the authors of the original studies on these datasets in the special issue. However, we believe our estimate is more robust than previous works, as it is based on well-measured and well-dated datasets, analyzed using a Bayesian multilevel model.

As a final remark, while the results provide the average of change points for the SCPs datasets, our findings for the $(^{239+240})Pu$ and $F^{14}C$ datasets demonstrate high precision. This precision is achieved by averaging multiple accurately dated and measured datasets. It is important to note that a singular sample inherently contains measurement inaccuracies or other uncertainties and biases. However, by employing a meta-analysis approach, particularly Bayesian (Kruschke and Liddell, 2018a; Kruschke

and Vanpaemel, 2015), we align closely with the concept of the “unknown gold standard” for research that seeks an overarching result (Ioannidis, 2005).

In essence, our multilevel errors-in-variables approach proves to be robust in identifying change points through regularization and averaging of multiple accurately measured datasets. This method will likely be useful in delineating other chronostratigraphic boundaries, especially those with associated uncertainty levels, such as the Pleistocene-Holocene boundary, where well-defined time series exist. Moreover, the philosophical approach underpinning this method, which emphasizes the use of all available data, provides valuable insights for defining any chronostratigraphic boundary.

Conclusions

The methodological requirements of the ICS for defining a stratigraphic boundary date involve selecting it from a single site. However, while one specific site may ultimately serve as the GSSP, we suggest that incorporating all available, relevant, and high-quality data through a robust statistical approach could greatly refine the determination of the transition date and its associated uncertainty. Such an approach may even support identifying the most representative GSSP site in the future.

Although the Anthropocene Epoch’s inclusion in the ICC has recently been rejected, we assert that our approach—integrating all available, relevant, well-measured, and dated datasets along with their uncertainties, and employing a meta-analysis technique—offers both practical and philosophical insights for defining the onset of the Anthropocene, should it be formally recognized in the future. This method, utilizing a multilevel Bayesian change point model with an errors-in-variables approach, aims to minimize potential flaws and biases inherent in individual datasets and to identify a plausible onset date for the Anthropocene Epoch.

Our key findings are as follows:

1. Analysis of $^{(239+240)}\text{Pu}$ data from seven globally distributed sites revealed a population-level change point at 1953 (95% HDI: 1951.5–1954.5). The median date suggests a lag of a couple of months associated with global fallout. Consequently, 1953 CE, reflecting a global signal, may be a more representative choice than 1952 CE for defining the onset of the Anthropocene Epoch; nevertheless, 1952 CE remains relevant as it falls within the HDI.
2. The analysis of the SCPs datasets reveals distinct geographical clustering for the change points: 1939 CE for North America, 1947 CE for Europe and 1954 CE for Asia. These findings suggest geographical influences on SCPs variations, which align with prior observations.
3. The F^{14}C results show a narrower range compared to $^{(239+240)}\text{Pu}$ and SCPs, with median dates predominantly around 1960 CE. With these well-measured data, we reveal that the lag of atmospheric carbon deposition in marine and limnic sediments is approximately 5.5 to 9 years, with a median of approximately 8 years.

Finally, the method applied in this study offers a novel multilevel approach for detecting change points in multiple datasets influenced by both chronological and methodological uncertainties. This approach holds the potential to extend beyond the Anthropocene boundary, possibly guiding the determination of dates for other stratigraphic boundaries, such as the Pleistocene-Holocene boundary, which is observed through relatively well-measured time series. Moreover, the philosophical basis of defining the onset date of a stratigraphic boundary using the total available evidence is crucial, and similar approaches, particularly Bayesian methods, can be adapted to any boundary. Additionally, this method has also the potential to provide an effective means of regularizing datasets and imputing missing data, especially in cases with known geometrical patterns.

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English of the given text if necessary” and was subsequently reviewed and further edited by the authors of this study. We would like to thank the two anonymous reviewers of this article for their valuable feedback. Additionally, we are grateful to all the authors of the articles published in the special issue aimed at identifying the GSSP for openly publishing or sharing their data and associated uncertainties.

Declaration of conflicting interests

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Code and data availability

All the data and codes used to generate the results can be found at https://github.com/zboraoon/anthropocene_date.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. This conclusion stems from the conclusive vote held in 2023 by the AWG members, during which the options of 1950 CE and 1952 CE were considered (Waters et al., 2023a).
2. The first H-bomb detonation occurred on 1 November, 1952.

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